Safety in Numbers: How Weather Conditions Inform Olive Ridley Nesting Events

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Sea turtles are an iconic and charismatic family of marine reptiles widely recognized for their large shells. However, not all sea turtles have hardback shells. The leatherback sea turtle is the one living species of sea turtle belonging to the Dermochelyidae family that, as its name suggests, has leather-like skin covering small, bony plates across its back. The other six species of sea turtle belonging to the Cheloniidae family all have hard shells, consisting of a carapace along its back and a plastron along its underside (Bennett, 2018). Other than their shells, sea turtles are also uniquely adapted to survive in the open ocean through traits like enhanced thermal regulation through blood circulation, the ability to hold their breath for extended periods of time, strong flippers to help them swim, and keratin spikes in their throats that help them feed underwater (Bennett, 2018; Osterloff, 2025).

Sea turtles play a crucial ecological role in coastal and marine environments. In general, they help to regulate the health of seagrass and coral communities by grazing and controlling populations of sponges and jellyfish (Bennett, 2018). They are also economically and culturally significant, having been the focus of cultural traditions and mythology for centuries, including those of the Iroquois people in North America, the Moche people of Peru, the Seri Indians from the Gulf of California, the Ancient Mayans, and the Ancient Chinese (Bennett, 2018).

Literature Review

The Olive Ridley sea turtle (*Lepidochelys olivacea*) is a dominantly pelagic species globally distributed across the tropical regions of the Pacific, Atlantic, and Indian oceans, and is the most abundant of the world's sea turtles (NOAA, 2025). *L. olivacea* belongs to the Cheloniidae family and its common name comes from its green color and heart-shaped shell (NOAA, 2025).

Though they live in the open ocean, adult Olive Ridleys periodically aggregate along the coast to reproduce. *L. olivacea* can engage in solitary nesting behavior, but they are also well known for engaging in synchronous nesting behavior, also termed "arribada," on tropical beaches like those in Mexico, Costa Rica, and India (Bernardo and Plotkin, 2007, as cited in Fonseca et al., 2009). Arribada nesting behaviors are a key evolutionary strategy that promotes nest survival by overwhelming predators with sheer numbers, since satiated predators cannot continue to prey on other nests (Eckrich and Owens, 1995).

Despite its abundance and large range, *L. olivacea* was globally listed as a declining Vulnerable species by the International Union for the Conservation of Nature's (IUCN) Red List in, 2008, making it a target for conservation efforts. One common strategy to recover sea turtle populations is to protect nests and increase recruitment (Eckert et al., 1999) by addressing common coastal threats like predation and erosion (Marcovaldi, 2001).

Due to arribada nesting being a direct evolutionary response to predation, understanding arribada behavior over time may be a crucial component to inform *L. olivacea* conservation management. Currently, it is known that *L. olivacea* arribada nesting in Ostional, Costa Rica is responsive to rainfall-related environmental cues like low salinity, higher relative humidity, and lower nearshore current velocity (Bézy et al., 2020). Additionally, arribada abundance significantly increases during the rainy season in Ostional, though hatching success may decrease from increased erosion and microbial loads in nests affecting embryonic mortality (Valverde et al., 2012), underscoring the importance of understanding the link between weather and nesting patterns.

Methods

Data Source and Cleaning

The data used in this analysis come from the Costa Rica Ministry of Environment and comprise individual nesting records of *L. olivacea* on Ostional nesting shores from, 2019 to, 2025 (N = 2,071 raw records). Each record corresponds to a single turtle encounter and includes a date, an event type (solitary or arribada), and a free-text weather description. Data were cleaned and analyzed in R Statistical Software (v4.5.2; R Core Team, 2025) with the tidyverse (Wickham et al., 2019) and janitor packages.

Weather and event-type fields were entered manually by multiple observers and contained substantial inconsistency, including spelling variants (e.g., “stormy,” “stormy rain,” “med. rain”), language mixing (e.g., “nublado”), and incidental data-entry artifacts such as a tag number erroneously appended to one event-type entry. All string values were trimmed of leading and trailing whitespace and case-folded before matching. Free-text values were then standardized into four weather categories (Clear, Partly Cloudy, Cloudy/Foggy, and Storm/Rain/Lightning) and two event types (solitary and arribada) using a lookup-table approach. Records with unresolvable or missing weather or event-type values (n = 511) were excluded.

Unit of Analysis

Because arribada is by definition a synchronized, mass nesting event, individual turtle records from the same night are not independent observations: they share both the same outcome and the same recorded weather conditions. The appropriate unit of analysis is therefore the nesting night rather than the individual turtle. Records were aggregated by date, producing one row per nesting night (N = 320 nights). Where multiple weather readings were recorded on the same night, the modal category was used. This yielded 266 solitary nights and 54 arribada nights across the seven-year study period.

Statistical Analysis

A chi-square test of independence was used to assess the overall relationship between weather type and nesting event type across the 320 nesting nights. To determine whether solitary and arribada frequencies differed from expectation within each weather category specifically, a chi-square goodness-of-fit test was conducted for each weather type separately. The expected frequency for each cell was calculated as (row total x column total) / grand total, representing the count expected if event type were distributed in the same overall proportion as in the full sample and were independent of weather. Because several cells had expected counts below five, Fisher's exact test was additionally computed to corroborate each goodness-of-fit result.

A binomial logistic regression (solitary = 0, arribada = 1) was used to estimate the odds of an arribada night in each non-clear weather category relative to clear conditions, with clear weather as the reference category.

Results

Nesting Event Distribution by Weather Type

Weather measurements in the raw dataset varied in description but fell into four broad categories: Clear, Partly Cloudy, Cloudy/Foggy, and Storm/Rain/Lightning. The frequency of all nesting nights ($N = 320$) was counted by weather type and nesting event type and used to create a 2x4 contingency table (Table 1).

An overall chi-square test of independence indicated a marginally non-significant association between weather type and nesting event type, $\chi^2(3, N = 320) = 6.67$, $p = .083$.

Per-category goodness-of-fit tests, using the baseline-adjusted expected value (row total x column total / grand total), showed that none of the four weather categories departed significantly from the overall 83%/17% solitary/arribada split: clear, $\chi^2(1) = 1.08$, $p = .299$ (Fisher's exact $p = .223$); partly cloudy, $\chi^2(1) = 2.23$, $p = .136$ (Fisher's exact $p = .196$); cloudy/foggy, $\chi^2(1) = 0.22$, $p = .639$ (Fisher's exact $p = .549$); and storm/rain/lightning, $\chi^2(1) = 3.15$, $p = .076$ (Fisher's exact $p = .096$). The storm/rain/lightning category approached but did not reach the conventional $\alpha = .05$ threshold, with only 1 of 26 storm nights recorded as an arribada event against a baseline-adjusted expectation of 4.4.

Table 1

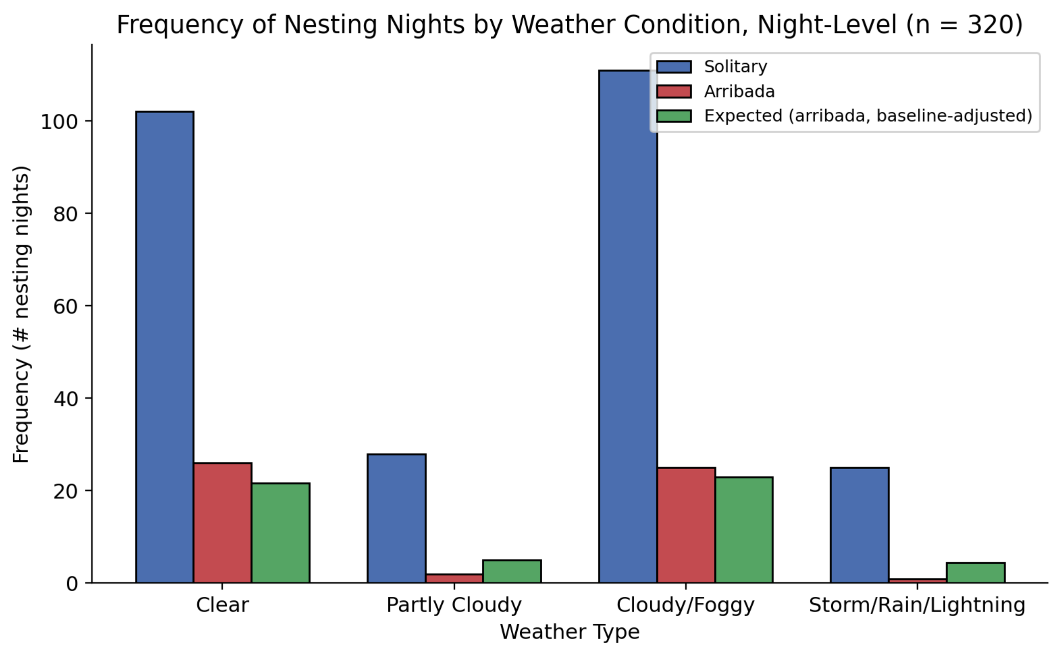
2x4 Contingency Table of Solitary and Arribada Nesting Night Frequency by Weather Type

Weather Type	Solitary	Arribada	Total	p (GOF)	p (Fisher)
Clear	102	26	128	.299	.223
Partly Cloudy	28	2	30	.136	.196
Cloudy/Foggy	111	25	136	.639	.549
Storm/Rain/Lightning	25	1	26	.076	.096
Total	266	54	320		

Note. Overall $\chi^2(3, N = 320) = 6.67$, $p = .083$. Each row represents one nesting night (modal weather and event type for that date). The p (GOF) column reports per-category goodness-of-fit tests using the baseline-adjusted expected value (row total x column total / grand total). Fisher's exact test is reported alongside because several expected cell counts are below five.

Figure 1

Bar Graph of Solitary and Arribada Nesting Night Frequency by Weather Type, with Baseline-Adjusted Expected Frequency



Note. This figure plots the frequency of solitary vs. arribada nesting nights across four weather categories (N = 320 nights). The green bar represents the baseline-adjusted expected count of arribada nights for each weather category, i.e., the count expected if nesting event type were distributed independently of weather in the same overall proportion as in the full sample (83% solitary, 17% arribada).

A binomial logistic regression confirmed the directional pattern suggested by the contingency table (Table 2). Relative to clear conditions, the odds of an arribada night were lower in partly cloudy conditions (OR = 0.28, 95% CI [0.06, 1.25], $p = .096$), cloudy/foggy conditions (OR = 0.88, 95% CI [0.48, 1.63], $p = .692$), and storm/rain/lightning conditions (OR = 0.16, 95% CI [0.02, 1.21], $p = .076$), though none of these contrasts reached conventional significance. The storm/rain/lightning estimate was the largest in magnitude and showed the widest confidence interval, reflecting the small number of storm nights in the sample ($n = 26$, only 1 of which was an arribada night).

Table 2

Odds Ratios of Arribada Nights for Non-Clear Weather Conditions

Predictor	**OR**	**2.5%**	**97.5%**	**p**
Clear (reference)	1.00	—	—	—
Partly Cloudy	0.28	0.06	1.25	.096
Cloudy/Foggy	0.88	0.48	1.63	.692
Storm/Rain/Lightning	0.16	0.02	1.21	.076

Note. OR = odds ratio of an arribada (vs. solitary) night relative to clear weather nights. CI = confidence interval. The intercept odds ratio (0.255) represents the baseline odds of an arribada night under clear conditions.

Discussion

The results of the chi-square and logistic regression tests indicate a directional pattern in which non-clear weather conditions are associated with lower odds of an arribada night relative to clear conditions, with the most consistent pattern observed for storm/rain/lightning conditions. This is especially true for inclement weather conditions, which showed the steepest reduction in arribada nesting: only 1 of 26 storm nights was an arribada night, against a baseline-adjusted expectation of 4.4. However, none of the tests reached the conventional $\alpha = .05$ significance threshold. The overall association between weather and nesting type was marginally non-significant ($p = .083$), and the most suggestive per-category result, storm/rain/lightning, reached $p = .076$ by chi-square goodness-of-fit and $p = .096$ by Fisher's exact test. This most likely reflects limited statistical power: the dataset contains only 54 independent arribada nights across the seven-year study period, which constrains the ability to detect a weather effect at conventional thresholds even when a true effect exists.

Because nesting event types are binary (arribada vs. solitary), this also implies that solitary events are more likely to occur during non-clear conditions. On the other hand, the goodness-of-fit test for clear conditions is not significant ($p = .299$), suggesting that other environmental factors might serve as stronger cues to trigger arribada nesting during clear conditions. This means that, although clear weather conditions do not single-handedly encourage arribada nesting, other weather conditions may deter these events, and this deterrence is most apparent for inclement conditions.

These results are particularly interesting in light of a previously noted paradox: if stormy conditions are associated with fewer arribada nights, why does the rainy season overall see significantly more arribada events than the dry season (Valverde et al., 2012)? This is surprising because if the relationship between rainy weather and arribada nesting frequency were truly negative, it would be reasonable to expect fewer arribada nesting events during times when rain is more frequent. Bézy et al. (2020) highlighted that low salinity, high humidity, and low ocean current velocity conditions are heavily associated with arribada events, which is further supported by Barik et al. (2014), who also found that low salinity and low energy oceanographic conditions are conducive to arribada nesting. One plausible explanation is that arribada nesting depends on a sequence of conditions rather than any single cue: recent storms generate low salinity and high humidity conditions, while a subsequent clear night ensures low current velocities. This sequence is more likely during the rainy season, when rain-to-clear transitions are common, than during the dry season, which would instead produce only clear conditions without the preceding storm signal. Bézy et al. (2020) also speculated that turtles may avoid nesting when short-term weather conditions make egg survival unlikely, including rainfall-induced erosion and increased microbial load (Valverde et al., 2012), further incentivizing emergence under clear conditions following a rainstorm.

Climate change may affect weather patterns to induce hotter and drier conditions and more extreme weather events (Castro-Vincenzi, 2024). If rainstorms are a significant cause of the low salinity and high humidity conditions that appear to precede arribada events, then a reduction in storm frequency could decrease the availability of these environmental cues, even if present-night storm conditions themselves deter nesting. So even though this analysis demonstrates that arribada nesting nights are less likely under stormy conditions, the reduction in upstream environmental cues caused by climate change may actually decrease overall arribada frequency. This information can be used to further push for climate resilience as a conservation management strategy, which may want to preserve arribada nesting behaviors as an anti-predation measure.

While there does not appear to be any one single cue that triggers arribada nesting, future studies may benefit from elucidating the potential for weather conditions to act as environmental cues, especially when climate change threatens to disrupt historic weather patterns. A more definitive test of the directional pattern observed here would be aided by an expanded sample of independently observed arribada nights, either through longer monitoring periods or pooling across additional arribada beaches. Directly measuring the environmental variables proposed as mechanistic cues (salinity, relative humidity, and nearshore current velocity) on and around each nesting night, and accounting for lunar phase, which is independently associated with arribada timing in the broader literature, would also strengthen future analyses.

Limitations

Several limitations of this study warrant acknowledgment. First, the effective sample size for inference is limited by the number of distinct arribada nights (n = 54), which constrains statistical power and likely explains why the directional patterns observed do not reach conventional significance thresholds. Second, weather conditions were recorded subjectively by multiple observers using free-text entries, introducing measurement variability that could not be fully corrected in post-hoc cleaning; standardized categorical weather recording in future data collection would improve consistency. Third, this analysis does not account for lunar phase, seasonality, or year-to-year variation in arribada frequency, all of which may confound the observed weather-nesting relationship. Fourth, the modal-aggregation approach used to assign a single weather category to each nesting night is a simplification, given that 126 of 320 nights had more than one weather reading recorded. Finally, the observational design precludes causal inference; the associations reported here are consistent with weather acting as an environmental cue for nesting behavior, but do not establish a causal mechanism.

Conclusion

This study examined whether present-night weather conditions predict the likelihood of an olive ridley arribada nesting event at Ostional, Costa Rica, using nesting night as the unit of analysis across 320 nights recorded between 2019 and 2025. The results suggest a directional pattern in which non-clear weather conditions, and stormy conditions in particular, are associated with lower odds of an arribada night relative to clear conditions. However, none of the statistical tests reached the conventional significance threshold, either because the dataset contains only 54 independently observed arribada nights, limiting the power available to confirm the pattern statistically, or because some number of explanatory variables are missing from the analysis.

These findings are consistent with a broader body of literature suggesting that arribada nesting in Ostional is shaped by a sequence of environmental cues rather than any single present-night condition. Storms may deter emergence on the night they occur while simultaneously generating the low salinity and high humidity conditions that precede and encourage arribada events in subsequent nights. This sequential relationship would explain why inclement weather suppresses arribada nesting locally on a given night while the rainy season as a whole remains the peak period for arribada activity.

The conservation implications of this work point toward climate resilience as a management priority. If reduced storm frequency under future climate scenarios diminishes the upstream environmental cues that appear to initiate arribada behavior, overall arribada frequency may decline even in the absence of direct human disturbance to nesting beaches. Continued long-term, night-level monitoring of both weather conditions and nesting behavior at Ostional, supplemented by direct measurement of salinity, humidity, and current velocity, would allow future studies to test this hypothesis more directly and to provide the sample of arribada nights needed to detect weather effects at conventional statistical thresholds.

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